Introduction to Econometrics and Linear Regression

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Chapter 1 –
Introduction to Econometrics and Linear Regression

Objectives

- Understand what Econometrics is and its importance
- Learn about different types of Econometric data
- Explore key modeling techniques
- Understand hypothesis testing in Econometric models
- Master the fundamentals of Linear Regression
- Learn about Ordinary Least Squares (OLS) estimation

What is Econometrics?

Econometrics integrates:

- Economic theory
- Mathematical modeling
- Statistical inference

Purpose: To analyze real-world economic data and understand complex economic phenomena.

Applications:

- Consumer behavior
- Economic growth
- Inflation analysis
- Market dynamics

Key Methods in Econometrics

Core Techniques:

- Regression Analysis - Exploring relationships between variables
- Time Series Analysis - Examining trends over time
- Panel Data Analysis - Combining cross-sectional and time dimensions
- Causal Inference - Identifying cause-and-effect relationships

Historical Development

History of Econometrics

Year	Contributor	Contribution
1926	Frisch	Coined the term “Econometric”
1936-37	Tinbergen	First aggregate-level Econometric model
1969	Klein	Economic forecasting models
1971	Kuznets	Pioneered GDP concept

Steps in Econometric Analysis

- **Step 1: Define the Problem:** Formulate a research question
- **Step 2: Collect Data:** Gather relevant economic data
- **Step 3: Define Hypotheses :** Based on existing economic theories
- **Step 4: Model Specification:** Choose appropriate mathematical model
- **Step 5: Estimation:** Use statistical techniques to estimate parameters
- **Step 6: Inference:** Test hypotheses and draw conclusions

Example - Tesla Stock Price & Unemployment

Hypothesis: Higher unemployment leads to lower Tesla stock prices

Variables:

- Dependent Variable (Y): Tesla stock price
- Independent Variable (X): Unemployment Rate

Model:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Where:

- β_0 = Intercept
- β_1 = Slope
- ϵ = Error term

Why Use Econometric Models?

- **Test Economic Theories:** Challenge assumptions empirically
- **Quantify Relationships:** Measure how variables affect each other
- **Forecasting:** Predict future trends (sales, GDP, unemployment)
- **Policy Evaluation:** Assess impact of policies
- **Evidence-Based Decisions:** Guide business and government strategies

Types of Data - Time Series

Definition: Data collected over time

Characteristics:

- Records observations at regular intervals
- Shows trends, seasonality, cycles

Example:

- Daily stock prices
- Monthly unemployment rates
- Quarterly GDP

python

```
dates = pd.date_range(start='2023-01-01', periods=100, freq='D')
values = pd.Series(range(100))
```

Types of Data - Cross-Sectional

Definition: Data collected from many subjects at one specific point in time

Characteristics:

- No time dimension
- Multiple observations at same time

Example:

- Survey responses
- Country statistics for one year
- Company financial data at a single date

python

```
data = {
   'Individual': ['A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J'],
   'Value': [5, 7, 8, 5, 6, 7, 8, 7, 6, 5]}
```

Types of Data - Panel Data

Definition: Combines time series and cross-sectional data

Characteristics:

- Same subjects observed repeatedly over time
- More information than either alone

Example:

- Tracking same companies over several years
- Following individuals over time in a survey

Types of Data - Dummy Variables

Definition: Binary variables indicating presence/absence of a characteristic

Values:

- **1** = Presence of characteristic
- **0** = Absence of characteristic

Examples:

- Gender (1 = Male, 0 = Female)
- Policy implemented (1 = Yes, 0 = No)
- Recession (1 = Yes, 0 = No)

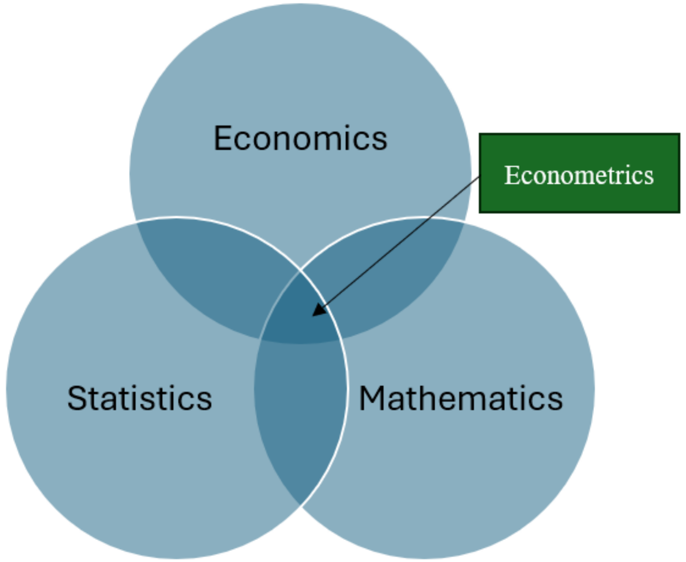
Economics vs Econometrics

Economics:

- Theoretical framework
- Studies how society allocates scarce resources
- Focus on “what should happen”

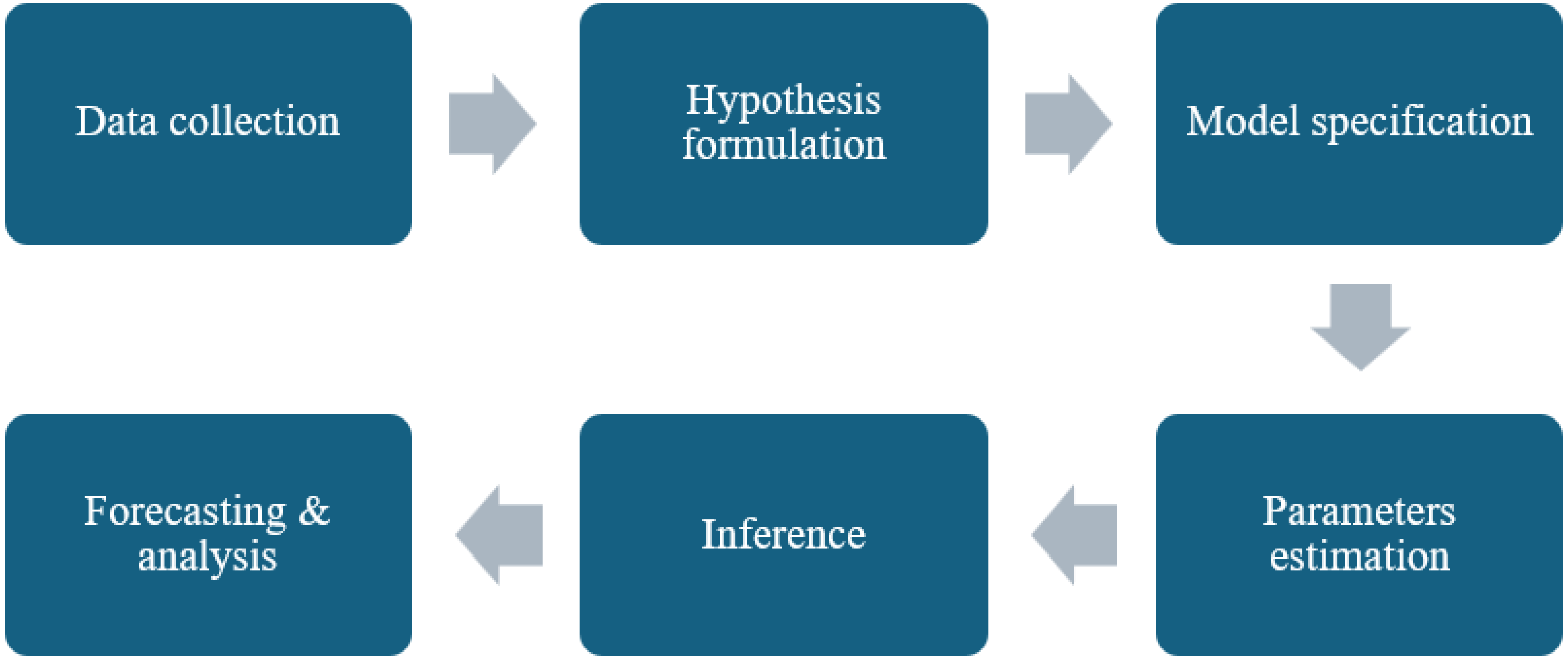
Econometrics:

- Quantitative analysis
- Uses statistics and mathematics
- Focus on “what actually happened”

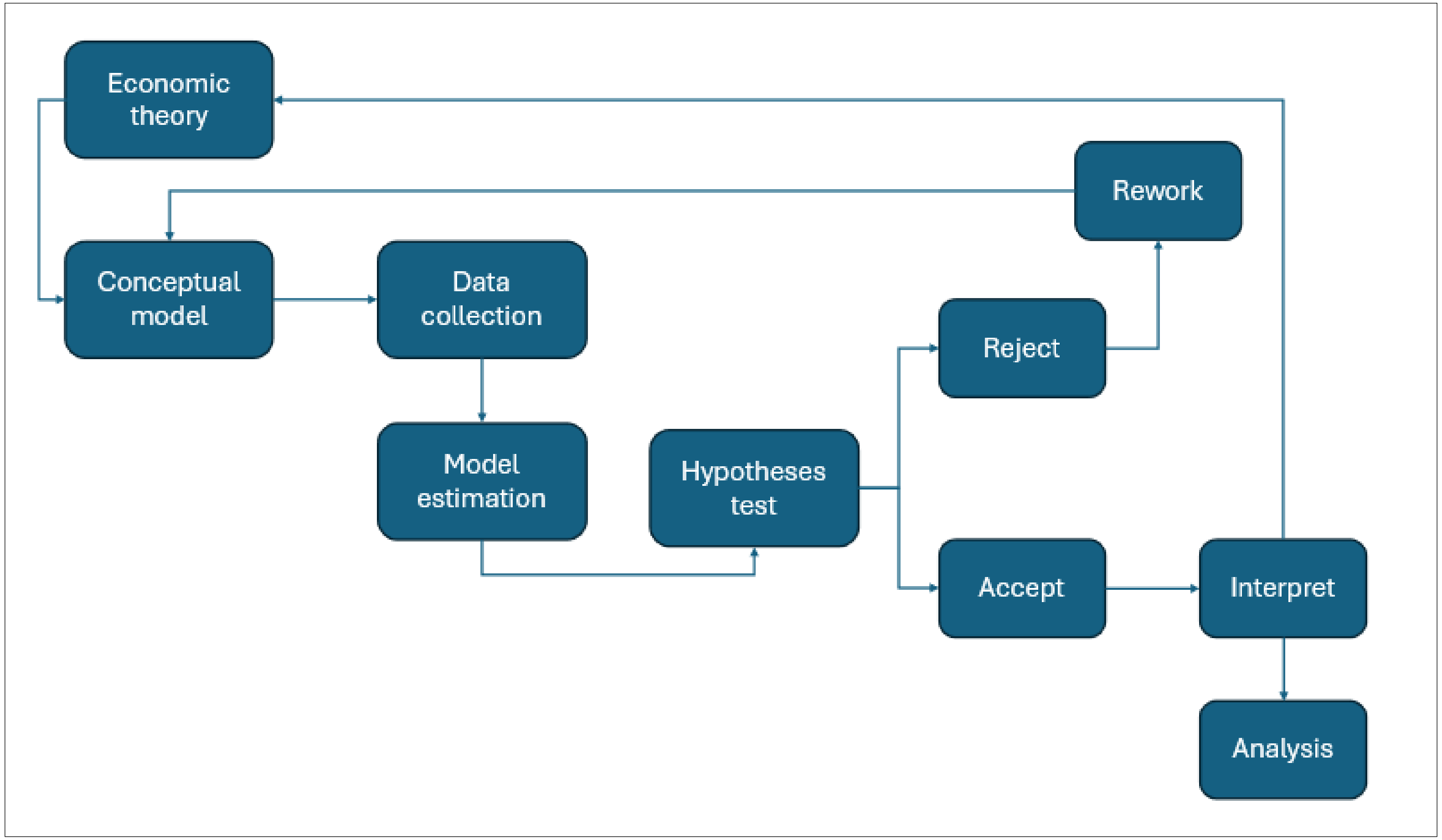


Economics	Statistics	Mathematics
Theory	Data analysis	Equations
Policy	Inference	Models
Behavior	Testing	Computation

Econometric Workflow



Modeling Steps



Limitations of Econometrics

Data Challenges

- Lack of data for testing theories
- Measurement errors in variables
- Data revisions that change results

Conceptual Limitations

- Focus on statistics over economic theory
- Correlation vs Causation confusion
- Spurious relationships without economic meaning

Practical Issues

- Small sample sizes in finance
- Low signal-to-noise ratio in financial data
- Model misspecification

Regression Analysis

Simple Linear Regression

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- One dependent variable
- One independent variable

Multiple Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \epsilon$$

- One dependent variable
- Two or more independent variables

Key Principle: Linear in Parameters

Linear Regression = Linear in parameters, not necessarily in variables

Examples of Linear Models:

- Exponential: $Y = \beta_0 + \beta_1 e^X + \epsilon$
- Logarithmic: $Y = \beta_0 + \beta_1 \log(X) + \epsilon$
- Polynomial: $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon$

Key Point: Coefficients (β) are not multiplied, divided, or raised to powers

Exponential Transformation

Model:

$$Y = \beta_0 + \beta_1 e^X + \epsilon$$

Interpretation:

- One-unit increase in X increases Y by:

$$\Delta Y = \beta_1 \left(e^X\right)(e - 1)$$

Characteristics:

- Accelerating growth as X increases
- Non-linear relationship in variables
- Linear relationship in parameters

Use Case: Modeling exponential growth processes

Logarithmic Transformation

Model:

$$Y = \beta_0 + \beta_1 \log(X) + \epsilon$$

Interpretation:

- β_1 = Elasticity of Y with respect to X
- 1% increase in X → β_1% change in Y
- Measures proportional relationships

Characteristics:

- Decelerating growth as X increases
- Stabilizes variance
- Reduces skewness

Use Case:

- Modeling percentage changes
- Economic elasticity analysis

Polynomial Transformation

Model:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon$$

Interpretation:

- β_1 = Linear effect (initial slope)
- β_2 = Quadratic effect (curvature)

Shape:

- $\beta_2 > 0$: Upward curve (accelerating)
- $\beta_2 < 0$: Downward curve (decelerating)

Use Case:

- Modeling non-linear relationships
- Capturing curvature in data

Importance of Regression in Econometrics

Factor	Description
Quantifying Relationships	Measure strength and direction between variables
Testing Theories	Validate economic hypotheses empirically
Forecasting	Predict future values from historical data
Policy Evaluation	Assess impact of policy measures
Causal Analysis	Identify cause-and-effect relationships
Handling Complexity	Analyze multiple variables simultaneously

Ordinary Least Squares (OLS)

Goal: Estimate parameters by minimizing Residual Sum of Squares (RSS)

$$RSS = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$$

Where:

- Y_i = Actual/observed value
- $\hat{Y}_i$ = Predicted value

Equation: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$

Why Minimize Squared Residuals?

Reason	Explanation
Minimize Errors	Ensures overall error is minimized
Equal Treatment	Both positive and negative deviations treated equally
Mathematical Simplicity	Quadratic optimization with easy solution
Statistical Properties	Unbiased, consistent, and efficient estimators
Gauss-Markov Theorem	Best linear unbiased estimators

Mathematical Derivation of OLS

Residual for observation i:

$$\mathrm{e} = \mathrm{Y} - (\beta_0 + \beta_1 X_i)$$

RSS to minimize:

$$\sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^{n} [Y_i - (\beta_0 + \beta_1 X_i)]^2$$

Solution: Set derivatives to zero:

$$\frac{\partial SSR}{\partial \beta_0} = 0, \qquad \frac{\partial SSR}{\partial \beta_1} = 0$$

Result: Solve normal equations for optimal $\hat{\beta}_0$ and $\hat{\beta}_1$

Key OLS Assumptions

- **Linearity**: Linear relationship between X and Y
- **Independence**: Observations are independent
- **Homoscedasticity**: Constant variance of errors
- **Normality**: Errors are normally distributed
- **No Multicollinearity**: Independent variables not correlated
- **Zero Conditional Mean**: Error term has mean zero

Log Transformation in Finance

Benefits:

- Linearize non-linear relationships
- Stabilize variance (heteroscedasticity)
- Reduce skewness in data
- Interpret as percentages/elasticity
- Compress large values, expand small ones

Growth Rate Approximation:

$$G_t = \frac{Y_t}{Y_{t-1}}$$

$$\log(1 + G_t) = \log\left(\frac{Y_t}{Y_{t-1}}\right) = \log(Y_t) - \log(Y_{t-1})$$

When is Log Approximation Reliable?

Taylor Expansion:

$$\log(1+X) = X - \frac{1}{2}X^2 + \frac{1}{3}X^3 - \cdots$$

For small growth rates (<10%):

$$\log(1+X) \approx X$$

Key Finding:

- < 10% growth: Log approximation is excellent
- > 10% growth: Error becomes noticeable
- Use division approach for large growth rates